

How to calculate p-values!!!

A coin is flipped twice and the outcome sequence is HEAD, HEAD. Is the coin fair?

When we flip a normal,
everyday coin, there's a
50% chance we'll get

Heads...

1st Flip

0.5

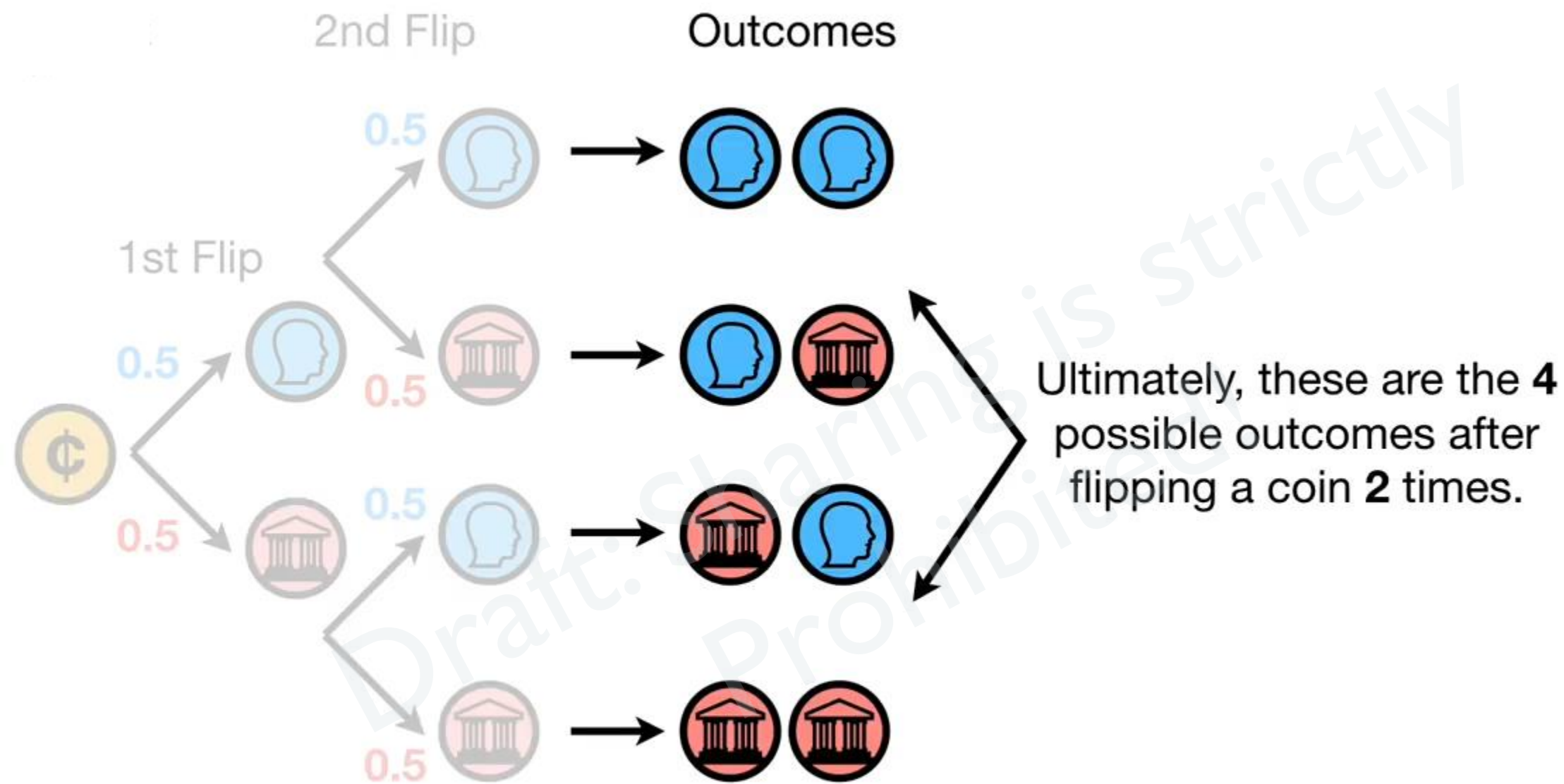


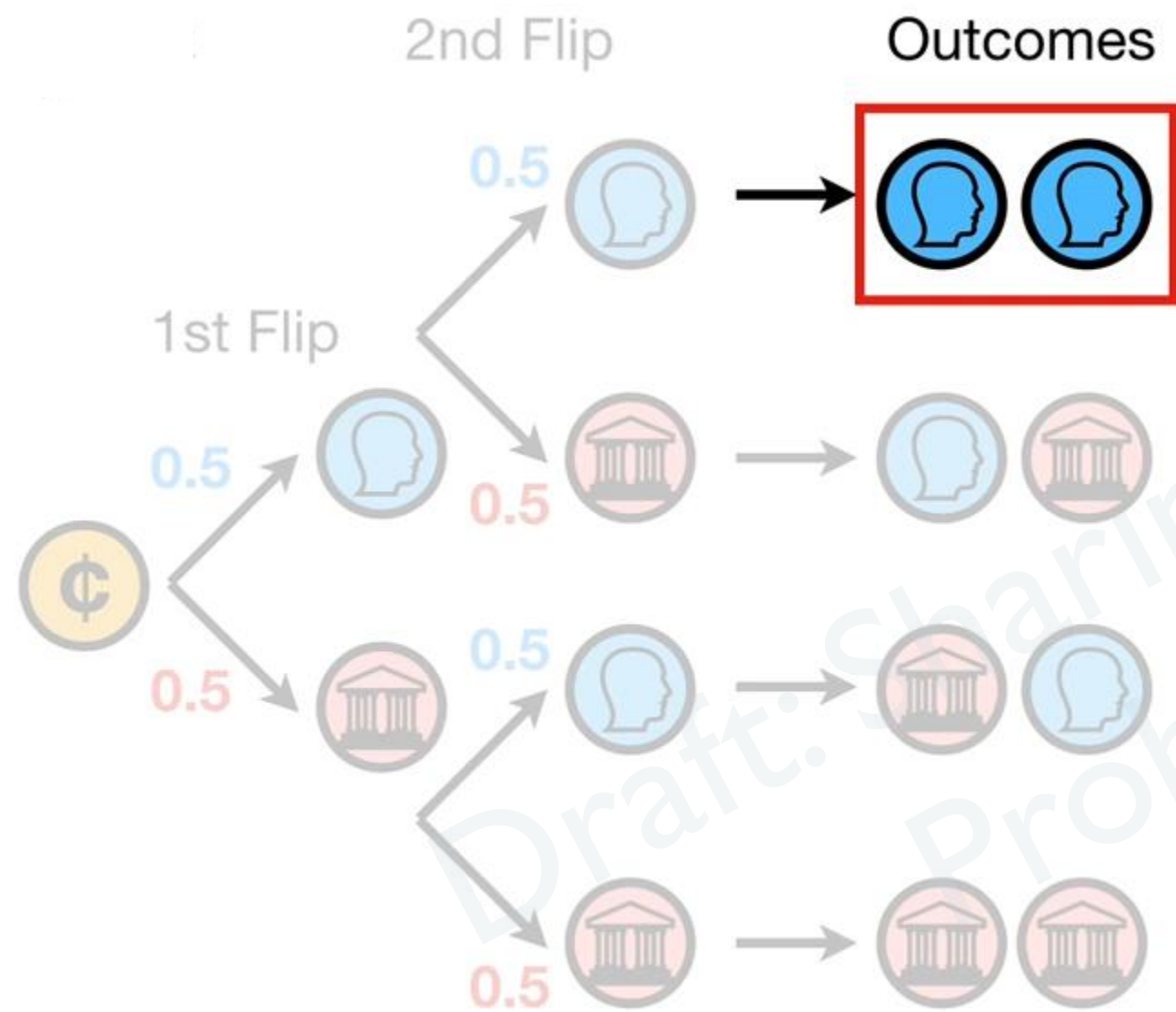
0.5



...and a **50%** chance we'll
get **Tails**.





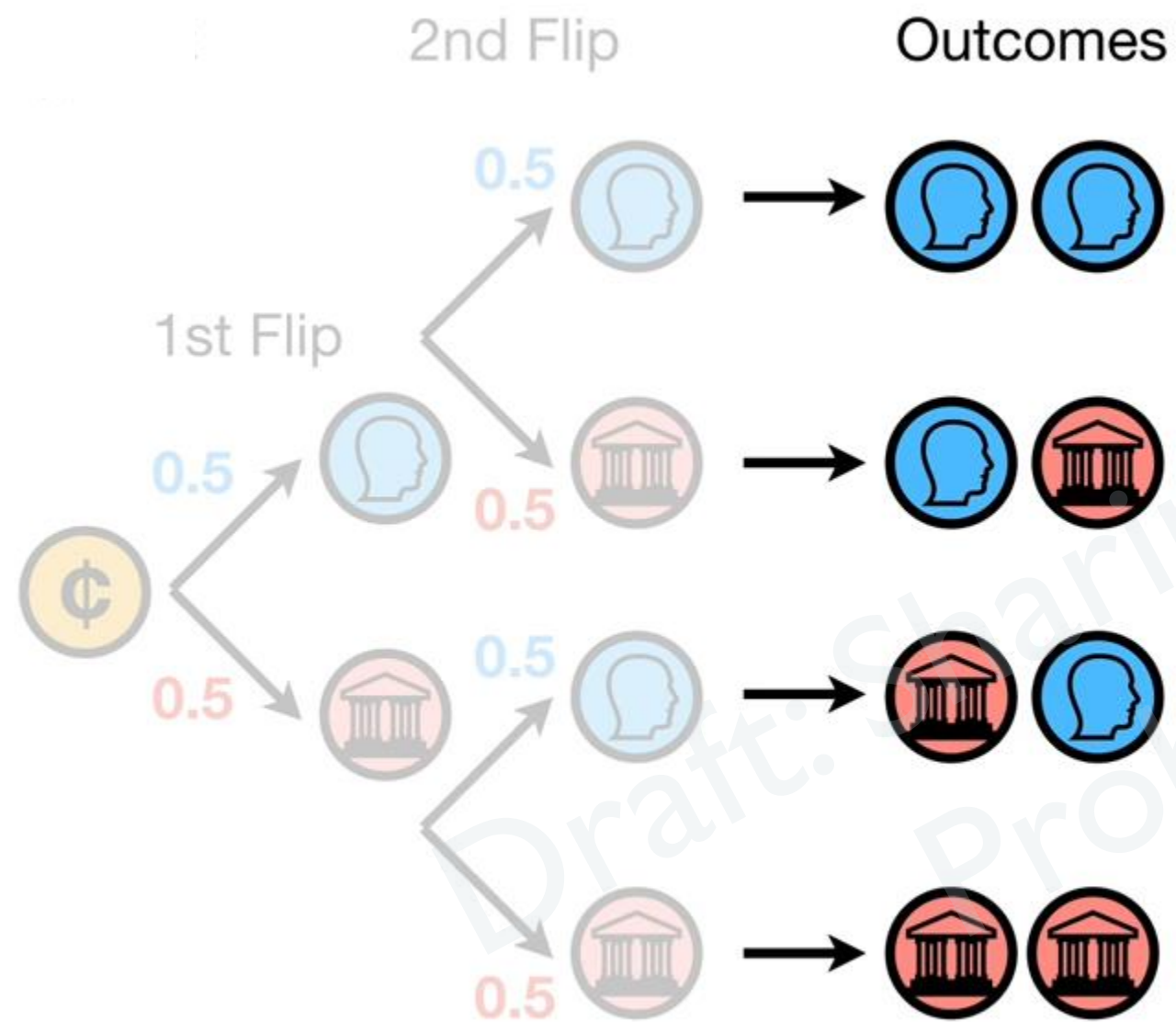


Thus, the *probability* of getting **2 Heads** is **0.25**.

The number of times
we got **2 Heads**.

The total number of
outcomes.

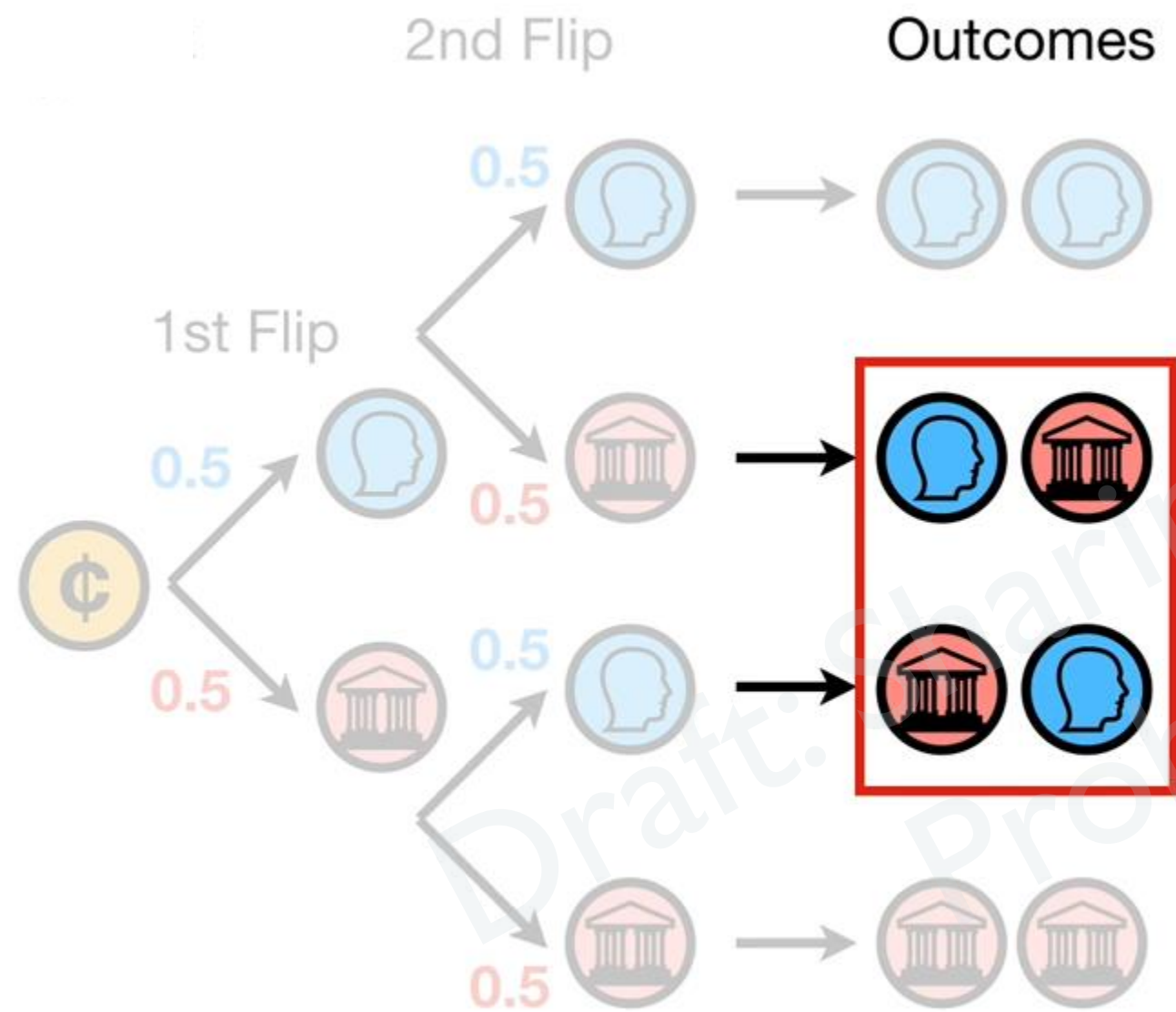
$$= \frac{1}{4} = 0.25$$



Likewise, the probability of getting **2 Tails** is...

The number of times
we got **2 Tails**.

$$\frac{\text{The number of times we got 2 Tails.}}{\text{The total number of outcomes.}} = \frac{1}{4} = 0.25$$



Finally, the probability of getting **1 Heads** and **1 Tails**, regardless of the order, is...

The number of times
Heads and **Tails**
occurred

$$\frac{\text{The total number of outcomes.}}{= \frac{2}{4} = 0.5}$$

p-value for **2 Heads** = $0.25 + 0.25 + 0 = 0.5$

1st Flip



2nd Flip



Outcomes Probability



0.25



0.5



0.25

p-value is the sum of three parts...

A **p-value** is composed of three parts:

- 1) The probability random chance would result in the observation.
- 2) The probability of observing something else that is equally rare.
- 3) The probability of observing something rarer or more extreme.

$$\text{p-value for 2 Heads} = 0.25 + 0.25 + 0 = 0.5$$

1st Flip



2nd Flip



Outcomes

Probability



0.25



0.5



0.25

Now the question is, **“Why do we care about things that are equally rare or more extreme?”**

A **p-value** is composed of three parts:

- 1) The probability random chance would result in the observation.
- 2) The probability of observing something else that is equally rare.
- 3) The probability of observing something rarer or more extreme.

$$\text{p-value for 2 Heads} = 0.25 + 0.25 + 0 = 0.5$$

1st Flip



2nd Flip



Outcomes

Probability



0.25



0.5



0.25

Because a lot of equally rare things would make something less special, we add **Part 2** to the **p-value**.

A **p-value** is composed of three parts:

- 1) The probability random chance would result in the observation.
- 2) The probability of observing something else that is equally rare.
- 3) The probability of observing something rarer or more extreme.

$$\text{p-value for 2 Heads} = 0.25 + 0.25 + 0 = 0.5$$

1st Flip



2nd Flip



Outcomes

Probability



0.25



0.5



0.25

And we add rarer things to the **p-value** for a similar reason.

A **p-value** is composed of three parts:

- 1) The probability random chance would result in the observation.
- 2) The probability of observing something else that is equally rare.
- 3) The probability of observing something rarer or more extreme.

OK, now that we know
that getting **2 Heads** in a
row is not very special or
statistically significant...



...what about getting 4
Heads and **1 Tails**?

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Prohibited!



The **p-value** for getting **4 Heads** and **1 Tails** is...

- 1) The probability we randomly get **4 Heads** and **1 Tails**:

$$\frac{5}{32}$$

Since **5** of the **32** outcomes had **4 Heads** and **1 Tails**.

HHHHH	THHHH	TTHHH	TTTTH
	HTHHH	THTHH	TTTHT
	THHHT	TTHHT	TTTHT
	HTTHH	THTTH	TTHTT
	HHTHH	HTHTT	TTHTT
	HTHHT	THHHT	TTHTT
	HTHHT	HTTTH	TTTTH
	HHHTT	HTTHT	TTTHT
	HHHTT	HTHTT	TTTHT
	HHHTT	HHTTT	TTTTT



The **p-value** for getting **4 Heads** and **1 Tails** is...

- 1) The probability we randomly get **4 Heads** and **1 Tails**:

$$\frac{5}{32} + \frac{5}{32}$$

- 2) The probability we randomly get something else that is equally rare:

Since **5** of the **32** outcomes has **1 Head** and **4 Tails**.

The diagram shows a grid of possible sequences (H/T) and highlights the most likely sequence in blue.

H	T	T	H	H	
H	T	T	H	T	H
H	T	T	H	H	T
H	T	H	T	T	H
H	T	H	T	H	T
H	T	H	H	T	T
H	T	H	T	T	H
H	T	H	T	H	T
H	T	H	T	T	T
H	T	H	H	T	T

The most likely sequence is highlighted in blue: **T T T H**



The **p-value** for getting **4 Heads** and **1 Tails** is...

- 1) The probability we randomly get **4 Heads** and **1 Tails**:

$$\frac{5}{32} + \frac{5}{32} +$$

- 2) The probability we randomly get something else that is equally rare:

- 3) The probability we randomly get something rarer or more extreme:



The **p-value** for getting **4 Heads** and **1 Tails** is...

- 1) The probability we randomly get **4 Heads** and **1 Tails**:

$$\frac{5}{32} + \frac{5}{32} + \frac{2}{32}$$

- 2) The probability we randomly get something else that is equally rare:

- 3) The probability we randomly get something rarer or more extreme:

HHHHH

Because both **5 Heads** and **5 Tails** only occurred once each, they are rarer than **4 Heads** and **1 Tails**.

TTTTT



The **p-value** for getting **4 Heads** and **1 Tails** is...

$$\frac{5}{32} + \frac{5}{32} + \frac{2}{32} = 0.375$$



Even though I got **4 Heads** and **1 Tails**, my coin is no different from a normal coin.

Again, we typically only reject the **Null Hypothesis** if the **p-value** is less than **0.05**...

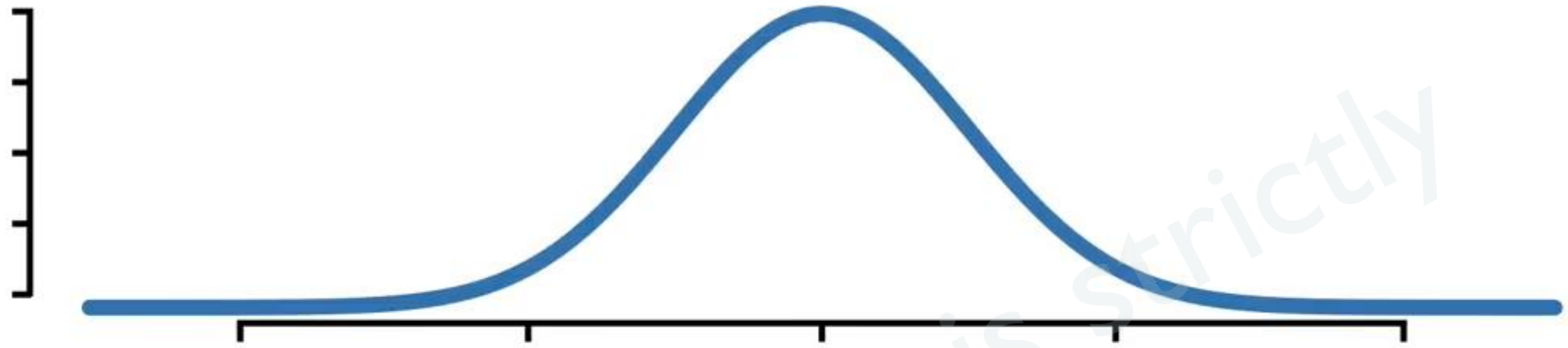
$$\frac{5}{32} + \frac{5}{32} + \frac{2}{32} = 0.375$$

But what if we wanted to calculate **probabilities** and **p-values** for how tall or short people are?

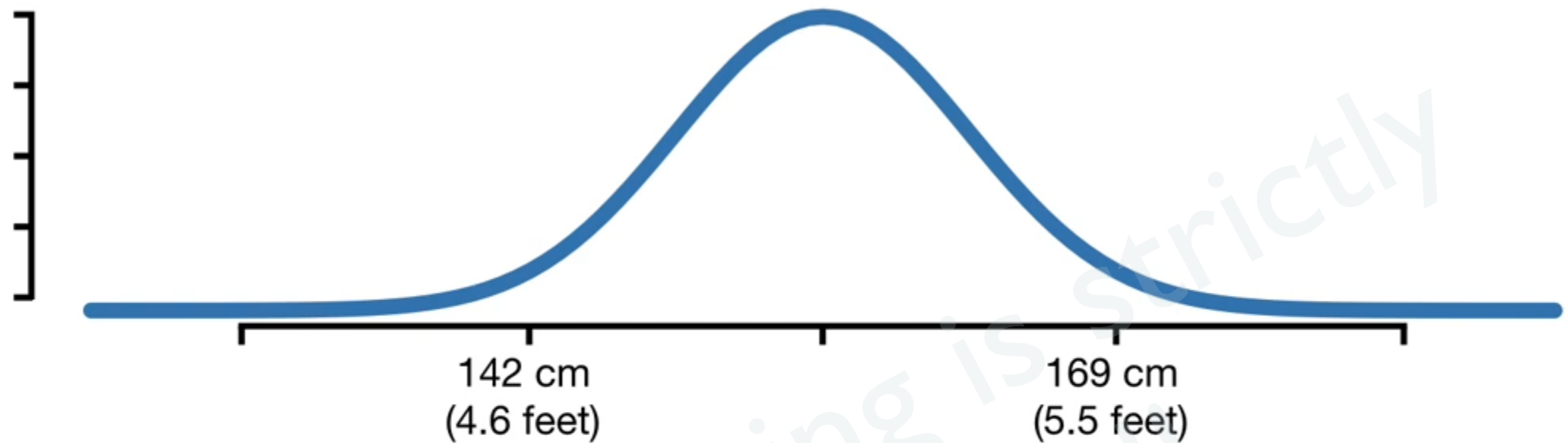


In theory, we could try to list every single possible value for height.

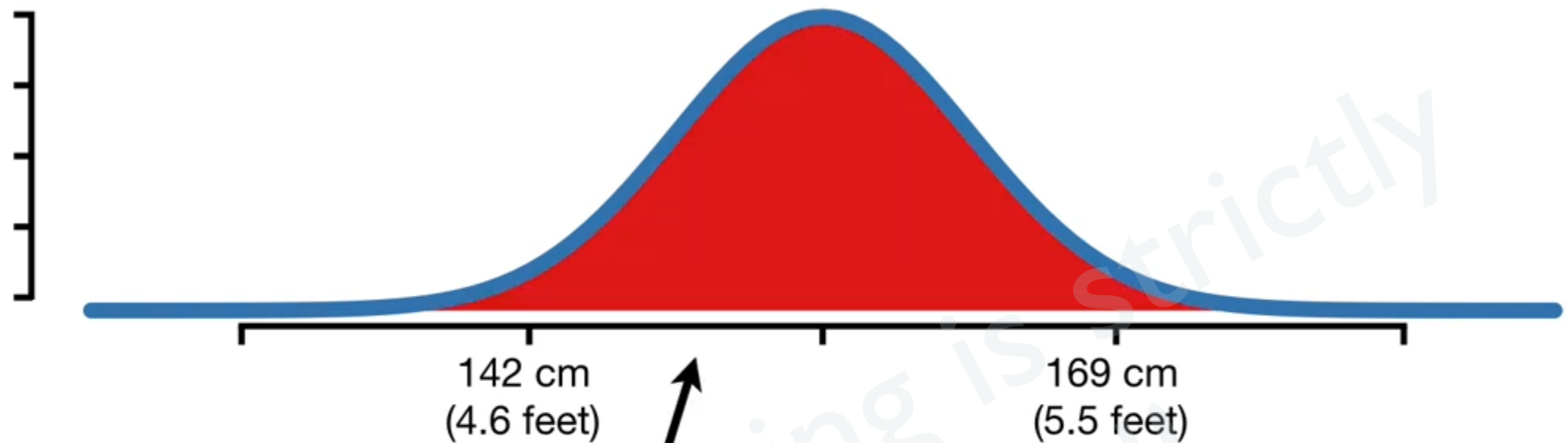
152.4 cm	152.9 cm	153.4 cm	etc...
152.5 cm	153.0 cm	153.5 cm	...
152.6 cm	153.1 cm	153.6 cm	etc...
152.7 cm	153.2 cm	153.6 cm	...
152.8 cm	153.3 cm	153.8 cm	etc...



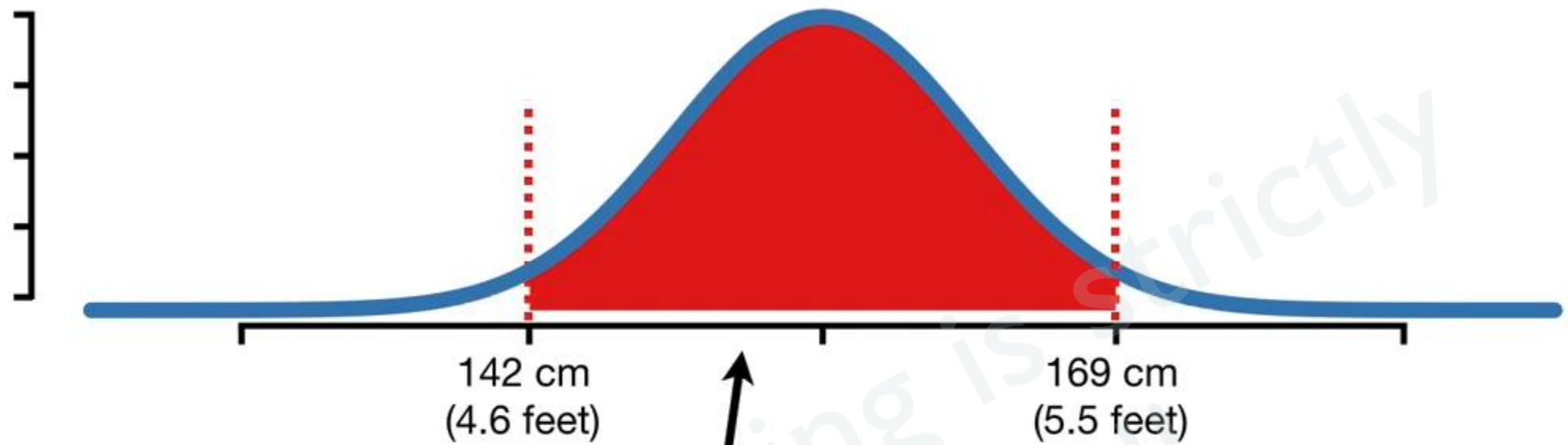
However, in practice, when we calculate **probabilities** and **p-values** for something continuous, like **Height**, we usually use something called a *statistical distribution*.



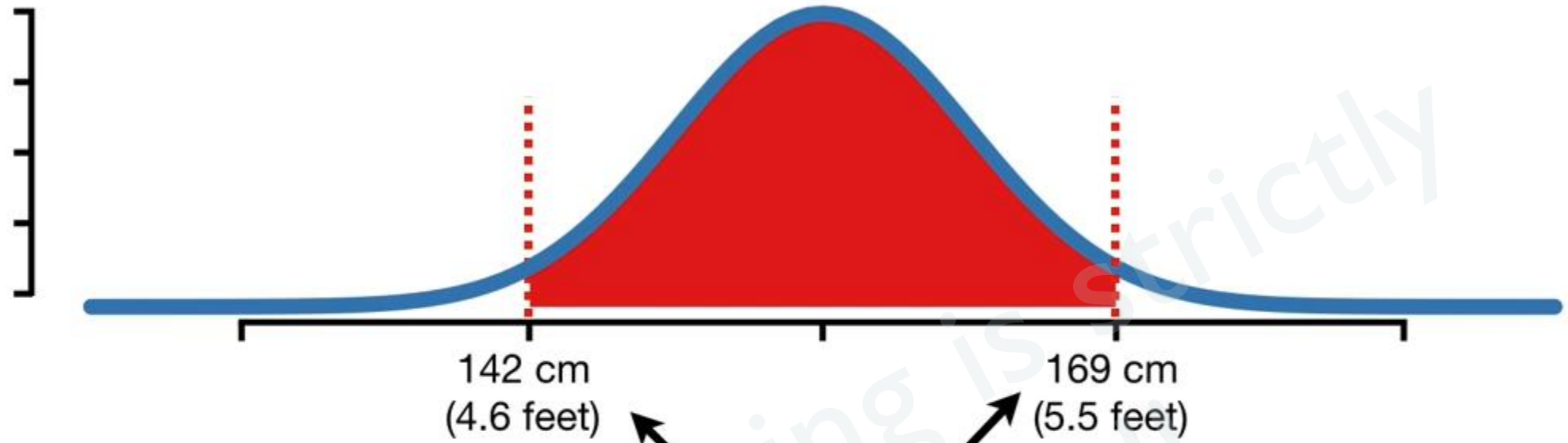
Here we have a distribution of height measurements from Brazilian women between **15** and **49** years old taken in **1996**.



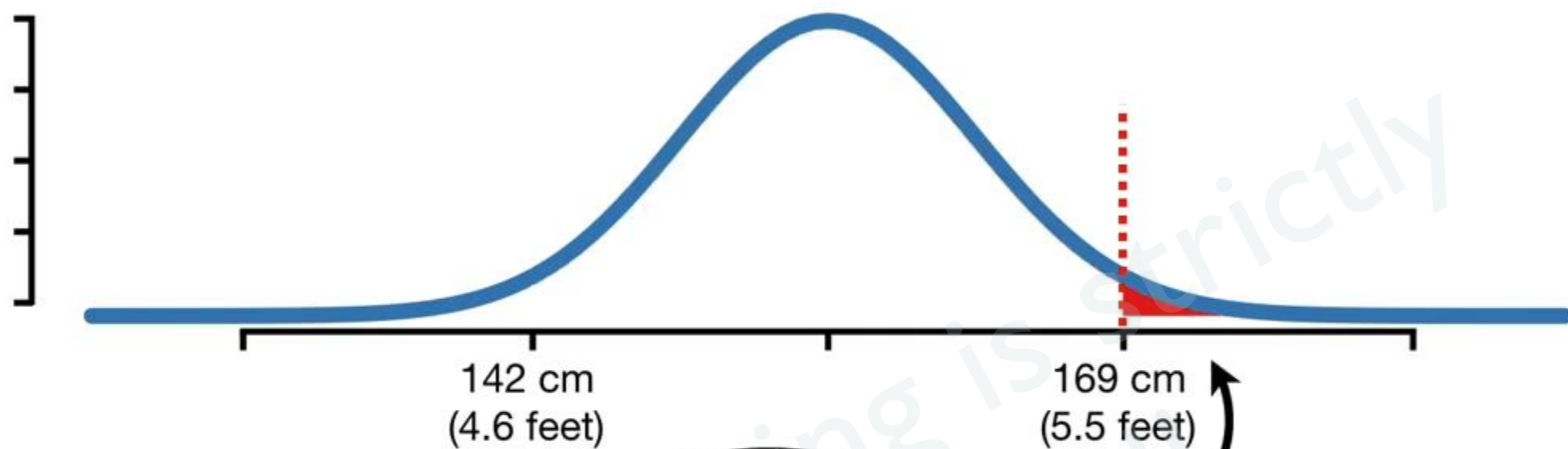
The **red area** under the curve indicates the probability that a person's height will be within a range of possible values.



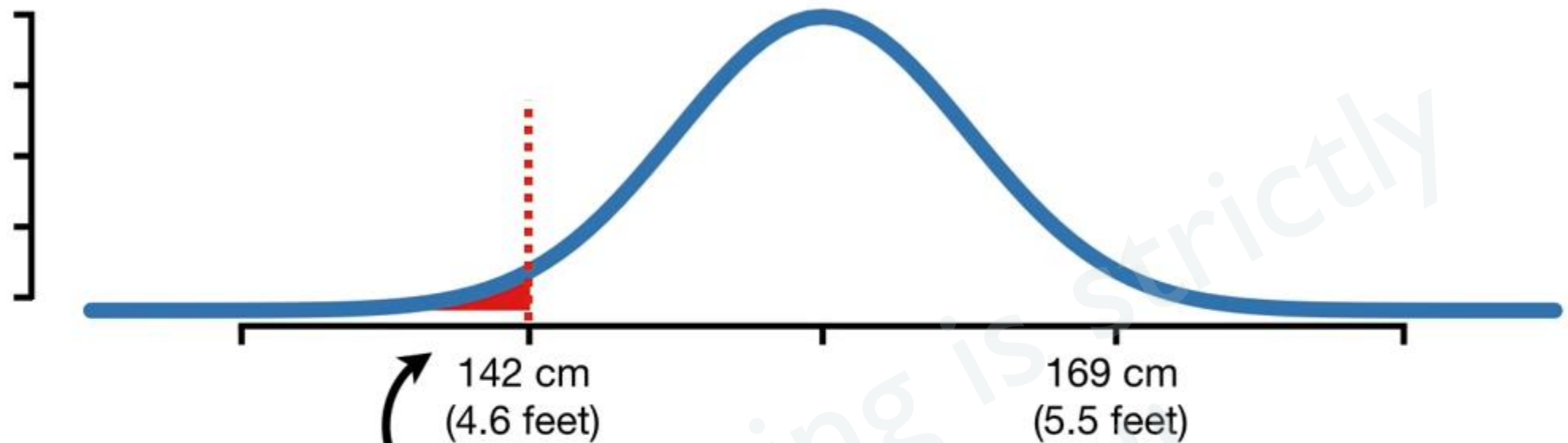
For example, **95%** of the area under the curve is between **142** and **169**...



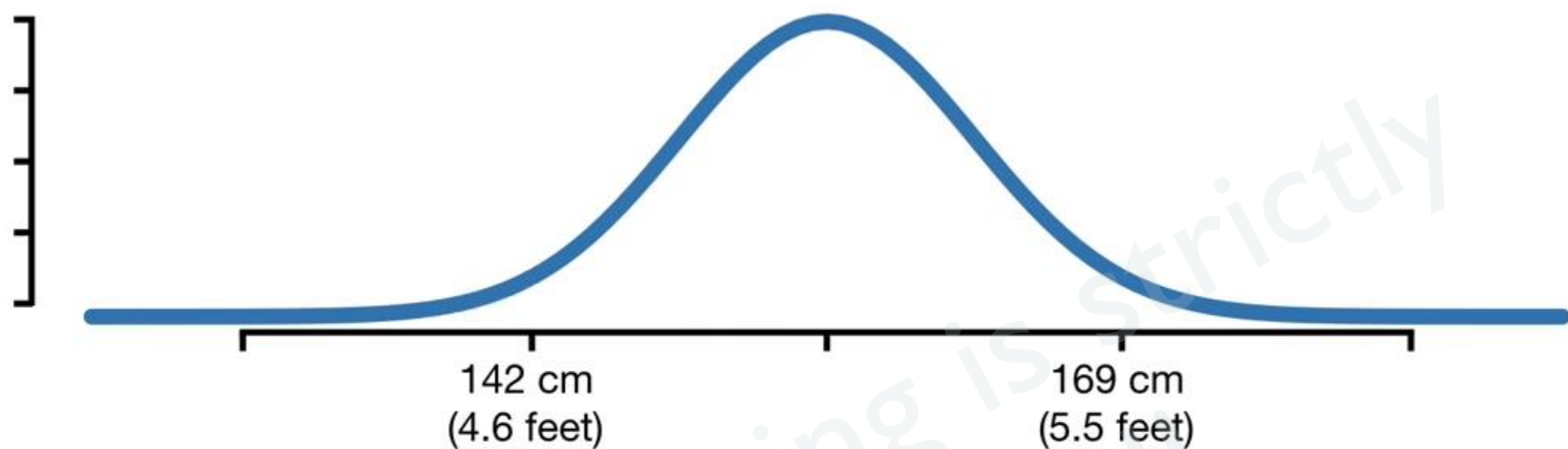
...and that means that **95%** of the Brazilian women were between **142** and **169** cm tall.



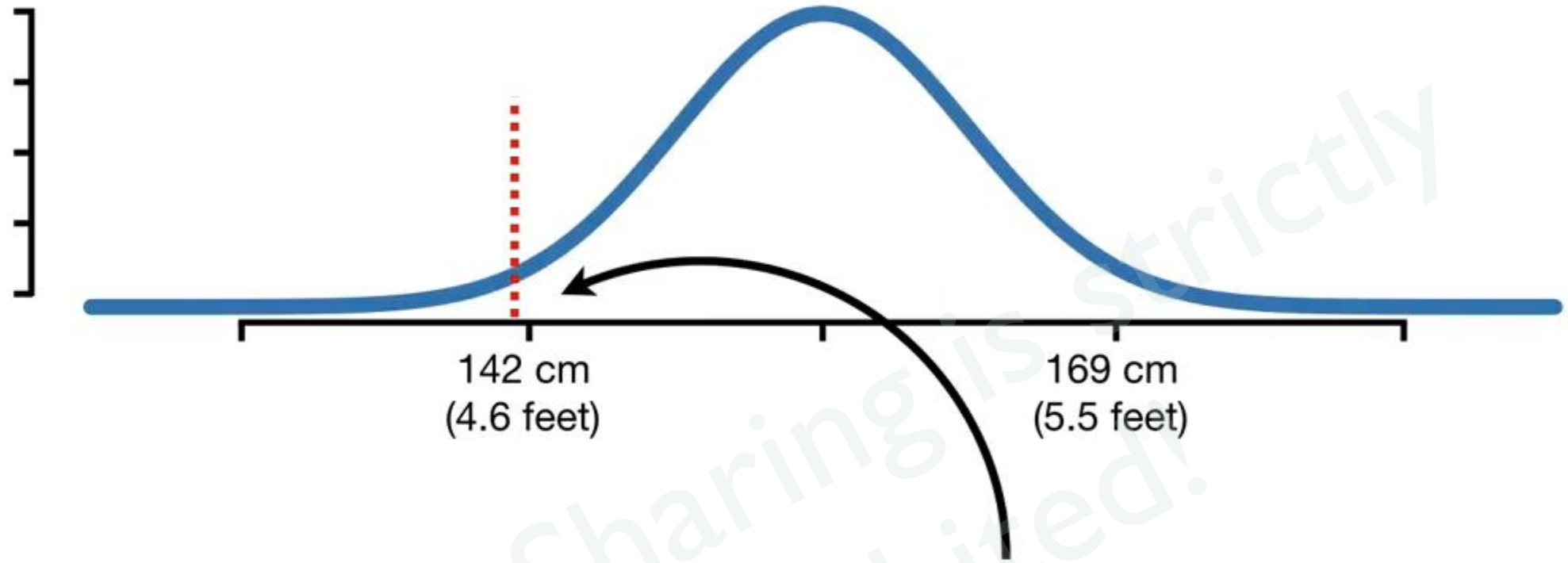
2.5% of the total area under the curve is greater than **169**.



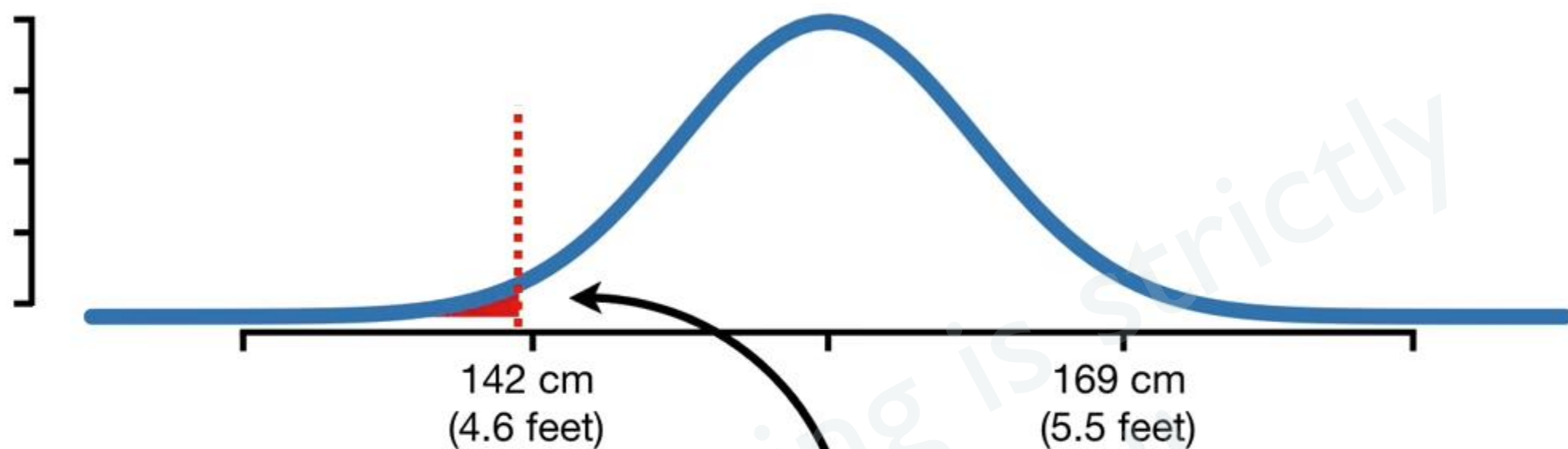
Likewise, **2.5%** of the total area under the curve is less than **142**.



To calculate **p-values** with a distribution, you add up the percentages of area under the curve.

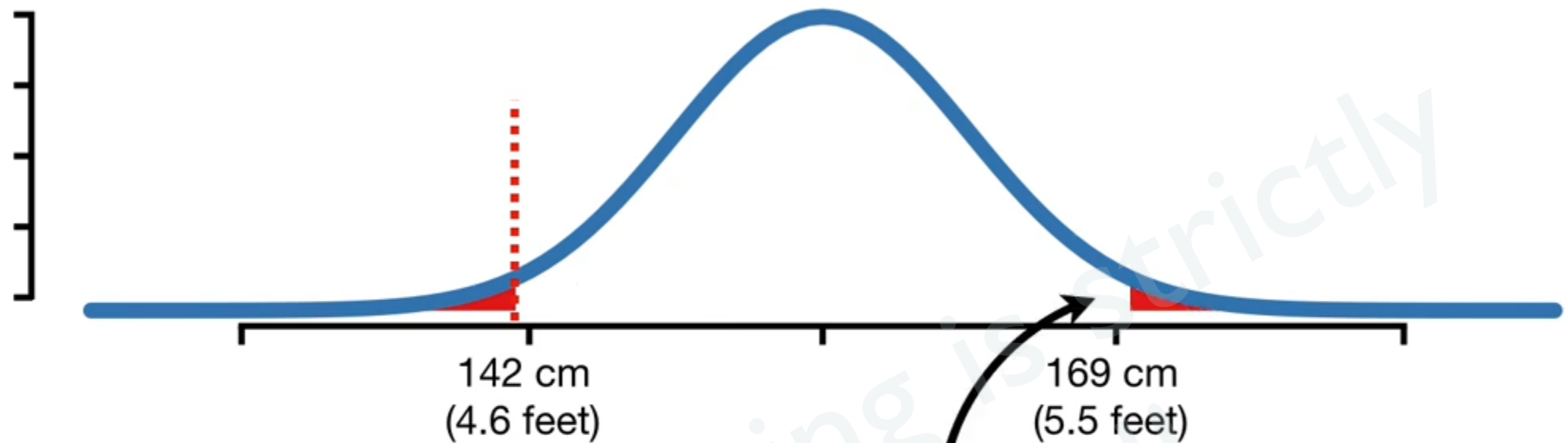


NOTE: If we had measured someone who was **141** cm tall, so just a little bit shorter than **142** cm...



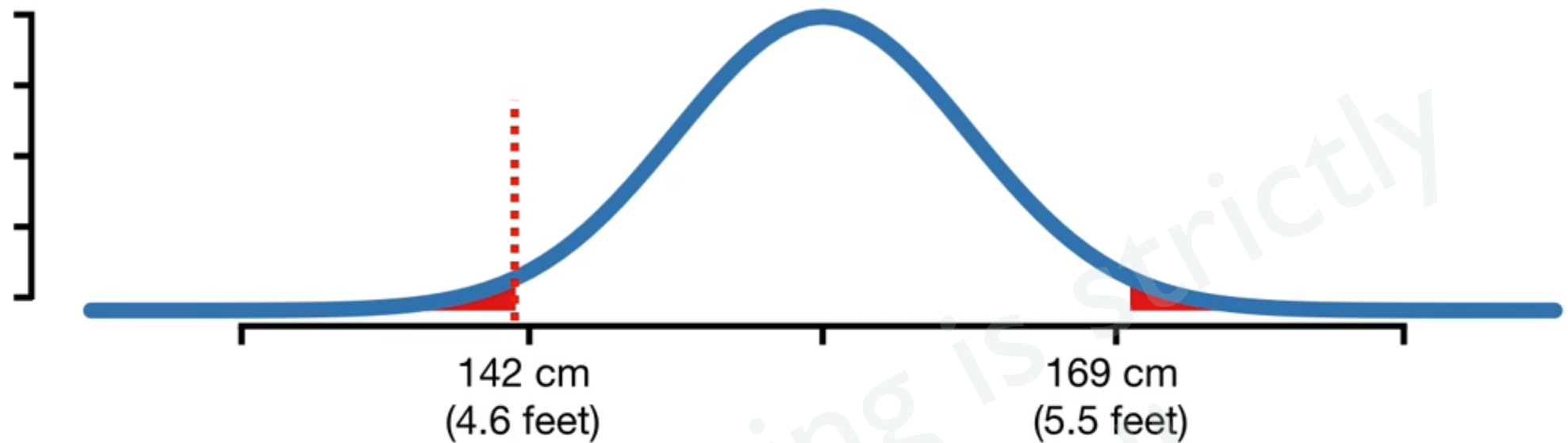
...then the **p-value**
would be **0.016**...

p-value for **141** cm given
the **blue distribution** = 0.016



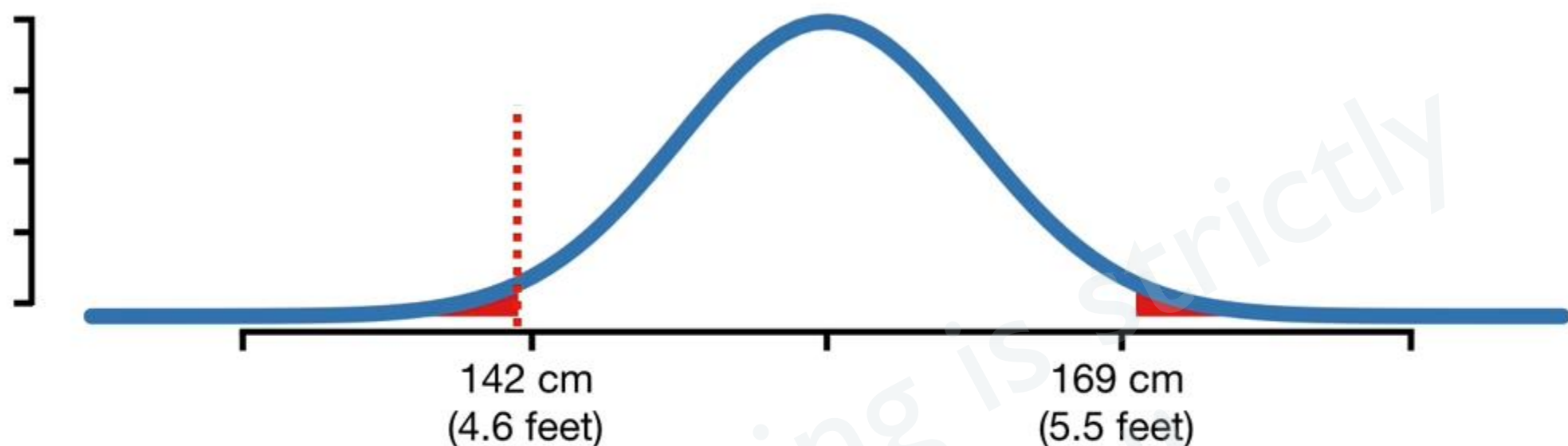
...plus **0.016...**

p-value for **141** cm given
the **blue distribution** = $0.016 + 0.016$



...which equals **0.03**.

p-value for **141** cm given
the **blue distribution** = $0.016 + 0.016 = 0.03$



And since **0.03 < 0.05**, the standard threshold, we can reject the hypothesis that, given the **blue distribution**, it is normal to measure someone **141** cm tall.

p-value for **141** cm given the **blue distribution** = $0.016 + 0.016 = 0.03$

THANK YOU!